

MUSANZE SAFE MARKET

UI AND UX DESIGN REPORT

SWE 3409 - Mobile Application Development | Assignment 1

Item	Information
Group	Group 13
Verification Code	MOB-G13-7341
Member 1	NZAYISENGA Erneste - 25/27341
Member 2	MANZI Cedrick - 25/27196
Member 3	INEZA Benitha - 25/27123
Member 4	KIRENGA Alain Thierry - 25/27792

1. Scenario

Our group designed a simple mobile application for a fictional market inspection pilot in Musanze. The application helps a field officer view assigned market stalls, enter one inspection, attach one evidence image, review the information and save the record during the current app session.

Problem Statement

A field officer needs a simple way to record stall inspection information using a phone. Paper notes can be difficult to organize and may miss important details such as category, risk level, consent or image evidence. Our application places these inspection steps in one mobile flow.

Target Field Officer

The main user is a field officer using an Android phone with Expo Go. The user needs clear navigation, readable text, simple forms and buttons that are easy to tap.

2. USER AND TASK FLOW

Primary User and Goal

Primary user: market field officer. The goal is to complete a fictional vendor inspection correctly, review the information and save the record.

Constraints

- No backend or database is required.
- All demonstration names and phone numbers are fictional.
- The interface should remain usable on a narrow Android phone.
- Permission denial and image picker cancellation must be handled.
- Invalid input must be blocked before review.

Complete Screen Flow

Home → New Inspection → Review → Records → Inspection Details

Back navigation from Inspection Details returns to Records without losing the current in-memory session.

Main Task Steps

1. View the market catalog on Home.
2. Open New Inspection and enter the required details.
3. Add an evidence image using Camera or Gallery.
4. Correct any field-level validation errors.
5. Review the validated information.
6. Save and view the record in Records.
7. Open Inspection Details for one saved record.

3. INFORMATION ARCHITECTURE

Tabs, Stack Routes and Data

Navigator	Route Name	Purpose	Data Passed / Used
Bottom Tab	Home	Market catalog and statistics	records, selected stall
Bottom Tab	New Inspection	Form and review	selected stall, form state
Bottom Tab	Records	Saved inspection list	records array
Stack	Inspection Details	One complete saved record	recordId

Main Data

- Market item data: stall code, name, category, zone, priority and icon.
- Inspection record: vendor alias, stall code, category, phone, risk, priority, consent, image and date/time.
- Review data: temporary validated information shown before saving.
- App state: the records array stays in memory while the app is running.

Navigation Choice

Bottom tabs make the three main areas easy to reach. The Records stack allows a saved record to open Inspection Details and then return to the list with Back.

4. LOW-FIDELITY WIREFRAMES



Figure 1: Home wireframe

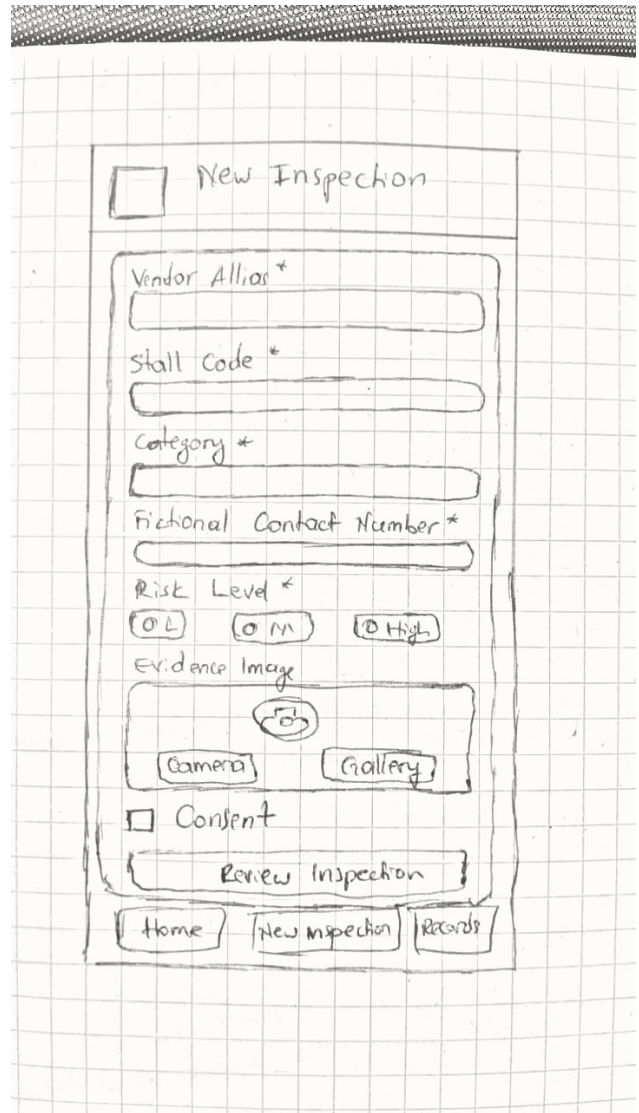


Figure 2: New Inspection wireframe

Wireframe Annotations

- Home: header, search, market stalls and bottom tabs.
- New Inspection: vendor alias, stall, category, contact number, risk level, priority, evidence image, consent and Review button.
- The group used simple paper sketches first, then implemented the same main screen structure in React Native.

5. REQUIRED STATES AND FORM FEEDBACK

Validation State

The screenshot shows a mobile app interface with a form for creating a new inspection. The form includes fields for a contact number, risk level, priority, and an inspection photo. The contact number field is highlighted in red with the error message "Exactly 10 digits - must start with 072, 073, 078, or 079" and "Contact number is required." Below the risk level and priority sections, there is a "Check Your Form" alert dialog box that says "Please correct the highlighted fields before submitting." The dialog box has an "OK" button. The form also includes a "Review Inspection" button at the bottom. The app's bottom navigation bar shows "Home", "New Inspection", and "Records" options.

Figure 3: field-level validation in the final implemented app

The form blocks invalid submission and displays messages close to the field that needs correction. The final app also shows a clear "Check Your Form" alert before the user can continue.

Other Required States

- Empty search state when no catalog item matches.
- Camera or Gallery selection state.
- Image preview with Replace and Remove actions.
- Permission denial and picker cancellation handling.
- Review state with validated fields, image, timestamp and group verification code.

6. VISUAL DESIGN

Color, Type, Spacing and Components

The application uses a simple beginner-level visual style. Green is used because it matches the Musanze Safe Market logo and safety theme. This academic report uses black text throughout.

Design Item	Choice	Reason
Header	Dark green (#1B5E20)	Matches the implemented app
Accent	Green (#2E7D32)	Active icons and buttons
Background	Light gray / white	Clean and readable
Type	Default React Native font	Simple for a beginner app
Spacing	Consistent padding	Avoids crowding
Components	Cards, chips, inputs, buttons	Reusable and easy to understand

Implemented Home Screen

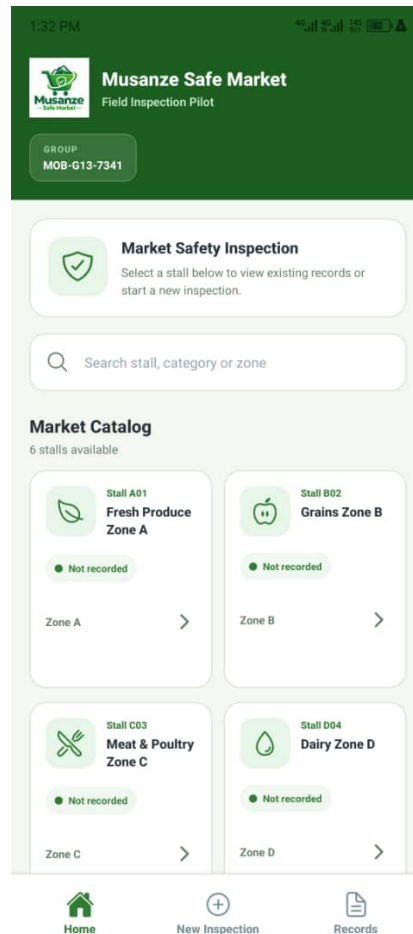


Figure 4: final Home screen with MOB-G13-7341

7. FINAL SCREEN MOCKUPS

Implemented Screens

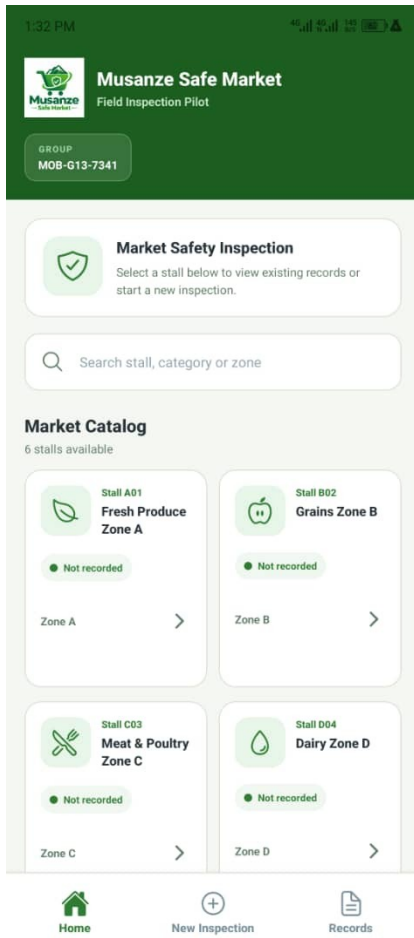


Figure 5: Home

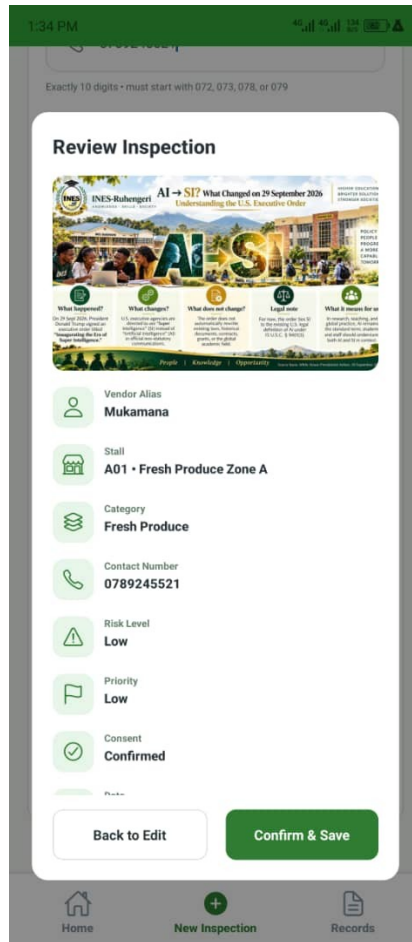


Figure 6: Review Inspection

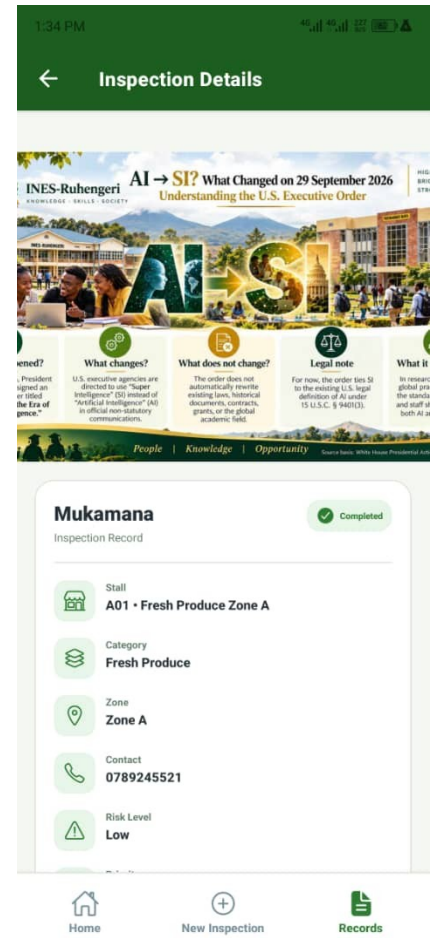


Figure 7: Inspection Details

These screenshots are from the final implemented application. They show the current layout, the correct verification code, review flow, selected evidence image and saved inspection details.

Additional Final Evidence

- The screenshots folder also contains the validation-error screen, empty Records state and saved Records list.
- The final screenshots use the same application version tested by the group on Android with Expo Go.
- The UI/UX report is kept readable at 100 percent zoom.

8. ACCESSIBILITY, TEAM RESPONSIBILITY AND SUBMISSION

Accessibility and Design Decisions

8. Contrast: dark text is used on light backgrounds, and white text is used on the dark green app header.
9. Touch targets: inputs and buttons have enough height and spacing for touch use.
10. Error messages: validation messages appear close to the field that needs correction.

Four-Member Team Roles

Member	Name	Role
1	NZAYISENGA Erneste	Product and UX Lead
2	KIRENGA Alain Thierry	Interface Engineer
3	INEZA Benitha	State and Navigation Engineer
4	MANZI Cedrick	Device Integration, QA and Release Evidence Lead

Moodle / Repository Information

Required Information	Final / Current Information
Group leader	NZAYISENGA Erneste - 25/27341
GitHub repository URL	https://github.com/nzemeste250/MOB_A1_G13
Current repository commit	4678771 - Update Group 13 app and testing evidence
Final commit hash	Update in README and Moodle after the final evidence commit
Demo video	Included in ZIP: evidence/MOB_A1_G13_DEMO.mp4

Submission Files

- MOB_A1_G13.zip - complete Expo project and supporting evidence.
- MOB_A1_G13_UIUX.pdf - this final 8-page UI/UX report.
- MOB_A1_G13_CONTRIBUTIONS.pdf - contribution and accountability record.